Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/10/2021
PI Name:	Swan, James Ph.D.		
Project Title:	Dynamic Self-assembly Driven by Time with Varying Fields		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	12/01/2017	**End Date:**	11/30/2020
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2020 per NSSC info (Ed., 11/12/19)		
Key Personnel Changes/Previous PI:	Final report February 2021: Emily Krucker continued to work on this project and is in her third year as a Ph.D. student. October 2019 report: The project entered its second year. Zachary Sherman completed his Ph.D. at MIT and a new second year graduate student, Emily Krucker began working on the proposed tasks. October 2018 report: The project began this year and a 4th year graduate student in chemical engineering at MIT, Zachary Sherman, started work on the proposed tasks.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0162		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is a research plan for using theory and simulation to analyze and extend the InSPACE 1, 2, 3, and 3+ investigations of the self-assembly of magnetically responsive nanoparticles in a time-varying magnetic field. The original work found that self-assembly was greatly accelerated by utilizing a simple open-loop control scheme of the magnetic interactions. This methodology offers one solution to the difficult engineering challenge of developing scalable and cost-effective fabrication methods for structured nanomaterials with proven applications in optoelectronics, energy storage and harvesting, biological sensing, 3D printing, and high-performance coatings. However, because of limited spatial and temporal resolution in the experiments, detailed analysis of the self-assembled structures was impossible. A deeper understanding of out-of-equilibrium self-assembly demonstrated in the InSPACE suite is needed before it can be leveraged to design processes that rapidly produce high quality nanomaterials. The proposed work has two main objectives. First, we will take advantage of insights developed since the InSPACE experiments for other out-of-equilibrium assemblies to re-evaluate the InSPACE data in the Physical Sciences Informatics (PSI) system. This will reveal new results that the original InSPACE team was unable to obtain. Second, we will use the InSPACE data in the PSI system to develop and validate a computational model of nanoparticle self-assembly in time-varying fields and use that model to extend the investigation of toggled magnetic field assembly beyond the parameter space explored in the experimental suite. This will aid in developing a predictive theoretical framework for dynamic self-assembly that can facilitate design of rapid nanomaterial syntheses. The work will utilize GPU-accelerated discrete element simulations of particle self-assembly driven by magnetic fields with detailed models of the magnetic and hydrodynamic interactions among the particles. The Principal Investigator's (PI) laboratory has recently developed new algorithms for modeling these interactions that have accelerated simulations by roughly a factor of 100 times. Consequently, discrete element simulations of the hierarchical self-assemblies observed in the InSPACE experiments are possible. The video micrographs from InSPACE are of limited spatial and temporal resolution; therefore, simulations capable of resolving the particle structure and dynamics on a local scale while replicating structure and growth kinetics on larger length scales and longer time scales provide additional insight into the InSPACE data. The ability to translate InSPACE microgravity data for the many-bodied, dynamic interactions among self-assembling particles into well-validated models for the same experiments will enable simulations to predict and then design dynamic self-assembly processes terrestrially as well as during continued space operations. The PI's laboratory has already demonstrated a predictive framework for other types of toggled self-assembly and is currently developing theories for the InSPACE assembly that can be used to predict the outcome and quantify the robustness of dynamical self-assembly processes. Incorporation of microgravity data into this theory is essential for testing its predictive capability. The PI was part of the original InSPACE 2 team as a post-doctoral scholar at the University of Delaware before starting as an assistant professor at Massachusetts Institute of Technology (MIT). He is first author of the two published papers on InSPACE experiments and is intimately familiar with the project. He has also published several papers extending toggled self-assembly to ground experiments and to simulations of new systems. One MIT graduate student, Zachary Sherman, currently completing a Ph.D. focused on toggled self-assembly processes will assist in the research. Their extensive experience makes the investigators uniquely qualified to re-examine and extend the InSPACE analysis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work focuses on far from equilibrium methods for assembling nano-particles. The goal is to discover routes to self assembly free of kinetic constraints introduced by equilibrium thermodynamics. If successful, these pathways will enable the production of microstructures materials with few defects at rates not accessible by current technological platforms. The potential implications for these materials technologies span advance manufacturing in the form of 3D printing, the development of high performance coatings, and the creation of functional or tunable photonics and phononic materials for micro-devices.
Task Progress:	During the most recent reporting period, we have worked to apply the simulations developed and validated against the InSpace data to the task of modeling colloidal systems relevant for terrestrial energy technologies. In this latest work, we have examined three key physical phenomena: 1. the way in which the ions in concentrated electrolytes structure in solution and how the effects the screening of electrostatic interactions among ions in solution, 2. the dynamics of ions in concentrated electrolytes and how collective ion motion leads to enhanced or suppressed conductivity of the electrolyte, 3. the interactions among charge colloidal particles in a concentrated electrolyte. The goal of these investigations has been to use the unprecedented ability to resolve complex interactions like electrostatics and fluid mechanics developed in our new simulation technologies to provide insights into important and long standing questions in this area. Measurements of how charge-charge interactions are screened in electrolytes have been made recently and do not match theoretical predictions. We've begun to explain why this is. The theory makes predictions of screening in the bulk while the experiments measure the interactions between surfaces near which the structure of the electrolyte is different. This work will continue after the award ceases as it has important implications for how to design battery technologies where controlling ion transport and particle interactions are critical engineering feats. During the extended period of performance a new student Emily Krucker was able to make considerable progress on this task while training to use the tools developed previously by the student Zachary Sherman. We are currently revising Emily's first peer reviewed manuscript and anticipate submission in the next few weeks. The completed work from this performance period should result in two other peer reviewed publications as well.
Bibliography Type:	Description: (Last Updated: 04/30/2025)
Articles in Peer-reviewed Journals	Sherman ZM, Swan JW. "Spontaneous electrokinetic magnus effect." Physical Review Letters. 2020 May 22;124(20):208002. https://doi.org/10.1103/PhysRevLett.124.208002 , May-2020

Articles in Peer-reviewed Journals	Krucker-Velasquez E, Swan JW. "Underscreening and hidden ion structures in large scale simulations of concentrated electrolytes." J Chem Phys. 2021 Oct 7;155(13):134903. https://doi.org/10.1063/5.0061230 ; PMID: 34624965 , Oct-2021
Articles in Peer-reviewed Journals	Wittmann L, Krucker-Velasquez E, Schaupp J, Westphal L, Swan JW, Alexander-Katz A, Bazant MZ, Schwaminger SP, Berensmeier S. "Influence of magnetic convection on separation efficiency in magnetophoretic microfluidic processes: a combined simulation and experimental study." Nanoscale. 2025;17(3):1574-84. http://dx.doi.org/10.1039/D4NR02225D , Jan-2025
Articles in Peer-reviewed Journals	Krucker-Velasquez E, Bazant MZ, Alexander-Katz A, Swan JW. "Dynamic response of concentrated electrolytes to chirp signals." ACS Nano. 2025 Apr 7;19(14):13673-84.https://doi.org/10.1021/acsnano.4c14099 , Apr-2025
Articles in Peer-reviewed Journals	Krucker-Velasquez E, Swan JW, Sherman Z. "Immersed boundary method for dynamic simulation of polarizable colloids of arbitrary shape in explicit ion electrolytes." J. Chem. Phys. 2024 Oct 21;161:164110. https://doi.org/10.1063/5.0224153 , Oct-2024